

Circuits and Systems: ECE 3364

Homework 1

This homework assignment includes problems from the textbook, which is Electric Circuits, by Nilsson and Riedel, 12th Edition. Please submit your solutions with all necessary work, through Canvas, by the due date.

Chapter 10: Problems 10.6, 10.8, 10.10, 10.24, 10.26

Note on units in Nilsson and Riedel textbook

In the Nilsson and Riedel textbook they are generally good about showing units, but there are some exceptions. Among the exceptions are that they assume that time is always given in units of [seconds] or [s]. So, when in Problem 10.6, when they tell us that

$$i_g = 30 \cos(25,000t) [\text{mA}],$$

what they mean is that

$$i_g(t) = 30 \cos\left(25,000 \left[\frac{\text{rad}}{\text{s}} \right] t\right) [\text{mA}], \text{ or } i_g(t) = 30 [\text{mA}] \cos\left(25,000 \left[\frac{\text{rad}}{\text{s}} \right] t\right).$$

Showing the (t) , which indicates the independent variable, is optional. The current units can be placed next to number it goes with, or at the end of the expression. I will show all units, but you are not required to show units where Nilsson and Reidel do not show those units. I place my units within square brackets, but you do not need to do this, either.

The second common exception is when labeling the units on certain dependent sources, specifically voltage-dependent current sources, and current-dependent voltage sources. This can be seen in Problem 10.8. The current dependent voltage source is that circuit is labeled

$$10i_\Delta,$$

And they assume that the units are [Volts/Ampere], or $[\Omega]$, so we would label that source as

$$10 \left[\frac{\text{V}}{\text{A}} \right] i_\Delta, \text{ or } 10 [\Omega] i_\Delta.$$

Again, I will show all units, but you are not required to show units where Nilsson and Reidel do not show those units.

In Problem 10.10, the line capacitor is labeled in an inappropriate way, in Figure P10.10. First, the diagram shows units for a variable. We do not do that; please ignore the units on the variable for the line capacitor. Next, the capacitor impedance is shown as $-jX$. The capacitor has a negative reactance, but we use X for reactance. Therefore, this is misleading, since the label is invariably the impedance of that component. In my solution, I solved for X as defined in the diagram, but gave my answer for reactance as the negative of my solutions for X . I believe that this problem should be modified.

Selected Numerical Solutions:

10.6. $990[\mu\text{W}]$

10.8. $2.125[\text{kW}]$

10.10. a) Reactance of the line is $-10[\Omega]$ or $-110[\Omega]$. b) For reactance of the line equal to $-10[\Omega]$, the $Q_{\text{del.by.source}} = 25[\text{kVAR}]$, which checks. For reactance of the line equal to $-110[\Omega]$, the $Q_{\text{del.by.source}} = -25[\text{kVAR}]$, which checks.

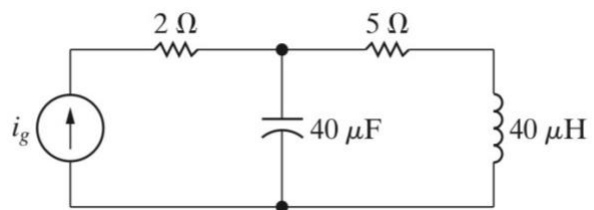
10.24. a) For Load 1, $\text{pf} = 0.9046$ lagging, and $\text{rf} = 0.4263$. For Load 2, $\text{pf} = 0.4346$ leading, and $\text{rf} = -0.9006$. For Load 3, $\text{pf} = 0.5740$ leading, and $\text{rf} = -0.81887$. b) For composite load, $\text{pf} = 0.9393$ lagging, and $\text{rf} = 0.34305$.

20.26. a) $P_{\text{del.by.Vg1}} = 4200[\text{W}]$, $Q_{\text{del.by.Vg1}} = 1200[\text{VAR}]$, $P_{\text{del.by.Vg2}} = 3600[\text{W}]$, $Q_{\text{del.by.Vg2}} = -300[\text{VAR}]$. b) $P_{\text{del.by.sources}} = 7800[\text{W}] = P_{\text{abs.by.loads}}$, which checks. $Q_{\text{del.by.sources}} = 900[\text{VAR}] = Q_{\text{abs.by.loads}}$, which checks.

10.6 Find the average power delivered by the ideal current source in the circuit in Fig. P10.6 if $i_g = 30 \cos 25,000t$ mA.

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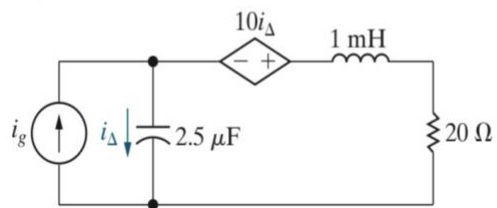
Figure P10.6



10.8 Find the average power dissipated in the $20\ \Omega$ resistor in the circuit seen in Fig. P10.8 if $i_g = 15 \cos 10,000t$ A.

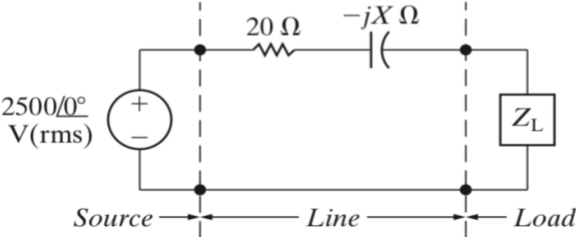
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Figure P10.8



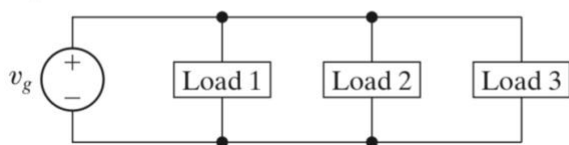
- 10.10** The load impedance in Fig. P10.10 absorbs 40 kW and 30 kVAR. The sinusoidal voltage source develops 50 kW.
- a) Find the values of capacitive line reactance that will satisfy these constraints.
 - b) For each value of line reactance found in (a), show that the magnetizing vars developed equals the magnetizing vars absorbed.

Figure P10.10



- 10.24** The three loads in the circuit in Fig. P10.24 can be described as follows: Load 1 is a $12\ \Omega$ resistor in series with a 15 mH inductor; Load 2 is a $16\ \mu\text{F}$ capacitor in series with an $80\ \Omega$ resistor; and Load 3 is a $400\ \Omega$ resistor in series with the parallel combination of a 20 H inductor and a $5\ \mu\text{F}$ capacitor. The frequency of the voltage source is 60 Hz .
- Give the power factor and reactive factor of each load.
 - Give the power factor and reactive factor of the composite load seen by the voltage source.

Figure P10.24





10.26 The three loads in the circuit seen in Fig. P10.26 are described as follows: Load 1 is absorbing 1.8 kW and 600 VAR; Load 2 is absorbing 1.5 kVA at a power factor of 0.8 leading; Load 3 is a $12\ \Omega$ resistor in parallel with an inductance whose reactance is $48\ \Omega$.

- a) Calculate the average power and the magnetizing reactive power delivered by each source if $\mathbf{V}_{g1} = \mathbf{V}_{g2} = 120\angle 0^\circ\text{ V (rms)}$.
- b) Check your calculations by showing your results are consistent with the requirements

$$\sum P_{\text{dev}} = \sum P_{\text{abs}};$$
$$\sum Q_{\text{dev}} = \sum Q_{\text{abs}}.$$

Figure P10.26

